Unit 5

Coordination Compounds

Modern inorganic and bio-inorganic chemistry, and much of the chemical industry, rest on coordination compounds.

What this unit sets out to do

- appreciate the postulates of Werner's theory of coordination compounds
- understand what is meant by: coordination entity; central atom or ion; ligand; coordination number; coordination sphere; coordination polyhedron; oxidation number; homoleptic; heteroleptic
- learn the naming rules for coordination compounds
- name mononuclear coordination compounds and write their formulas
- define the types of isomerism they show
- understand their bonding through Valence Bond and Crystal Field theories
- appreciate their importance and uses in daily life

From complexes to coordination compounds

- transition metals form many complex compounds in which the metal atom is bound, by electron sharing, to several anions or neutral molecules
- today these are called **coordination compounds**; their chemistry is an important and demanding part of modern inorganic chemistry
- new ideas of bonding and molecular structure explain how they work as vital parts of living systems

Where they matter

- magnesium, iron and cobalt form the coordination compounds chlorophyll, haemoglobin and vitamin B_{12}, in that order
- many metallurgical processes, industrial catalysts and analytical reagents use them
- also electroplating, textile dyeing and medicinal chemistry

5.1 Werner's Theory of Coordination Compounds

Alfred Werner (1866–1919)

- Swiss chemist, first to set out ideas on the structures of coordination compounds
- made and characterised large numbers of them, using simple experiments to probe how they behave physically and chemically
- proposed that a metal ion has a **primary valence** and a **secondary valence**

Evidence from cobalt(III) chloride–ammonia

- binary compounds $CrCl_3$, $CoCl_2$ and $PdCl_2$ have primary valences of 3, 2 and 2
- in a series of cobalt(III) chloride–ammonia compounds, excess cold silver nitrate precipitated some of the chloride as AgCl, while some stayed in solution

Moles of AgCl precipitated

1 mol of	Colour	AgCl formed
$CoCl_3.6NH_3$	yellow	3 mol
$CoCl_3.5NH_3$	purple	2 mol
$CoCl_3.4NH_3$	green	1 mol
$CoCl_3.4NH_3$	violet	1 mol

Interpretation

- these results, and the conductivities of the solutions, make sense if (i) cobalt keeps six groups attached throughout (chloride ions, ammonia molecules or a mixture), and (ii) the compounds are written as in Table 5.A, the square-bracketed atoms forming one unit that stays intact under the conditions of the reaction
- Werner called the number of groups bound directly to the metal ion its **secondary valence**; here it is six in every case

TABLE 5.A **Formulation of Cobalt(III) Chloride-Ammonia Complexes**

Colour	Formula	Solution conductivity matches
Yellow	$[Co(NH_3)_6]^{3+}3Cl^-$	1:3 electrolyte
Purple	$[CoCl(NH_3)_5]^{2+}2Cl^-$	1:2 electrolyte
Green	$[CoCl_2(NH_3)_4]^+Cl^-$	1:1 electrolyte
Violet	$[CoCl_2(NH_3)_4]^+Cl^-$	1:1 electrolyte

The last two compounds share the empirical formula $CoCl_3.4NH_3$ yet differ in properties: such compounds are **isomers**.

Werner's postulates (1898)

- 1. metals in coordination compounds show two kinds of linkage (valence): primary and secondary
- 2. primary valences are usually ionisable and are met by negative ions
- 3. secondary valences do not ionise and are met by negative ions or neutral molecules; a metal's secondary valence is fixed, and equals its coordination number
- 4. the ions or groups held by secondary linkages take up characteristic spatial arrangements for each coordination number

Modern terms

- these spatial arrangements are now called **coordination polyhedra**
- what lies within the square bracket is the **coordination entity** (or **complex**); ions left outside it are **counter ions**

Common shapes

- Werner also held that octahedral, tetrahedral and square planar shapes are the commonest for transition metal complexes
- $[Co(NH_3)_6]^{3+}$, $[CoCl(NH_3)_5]^{2+}$ and $[CoCl_2(NH_3)_4]^+$ are octahedral; $[Ni(CO)_4]$ is tetrahedral and $[PtCl_4]^{2-}$ square planar

Worked example secondary valences

From these observations on aqueous solutions, give the secondary valence of the metal in each compound.

Formula	Moles of AgCl precipitated per mole of compound with excess $AgNO_3$
(i) $PdCl_2.4NH_3$	2
(ii) $NiCl_2.6H_2O$	2
(iii) $PtCl_4.2HCl$	0
(iv) $CoCl_3.4NH_3$	1
(v) $PtCl_2.2NH_3$	0

- (i) secondary valence 4
- (ii) secondary valence 6
- (iii) secondary valence 6
- (iv) secondary valence 6
- (v) secondary valence 4

Double salt vs complex

- both form when two or more stable compounds combine in a fixed stoichiometric ratio
- double salts, e.g. carnallite $KCl.MgCl_2.6H_2O$, Mohr's salt $FeSO_4.(NH_4)_2SO_4.6H_2O$ and potash alum $KAl(SO_4)_2.12H_2O$, break up completely into simple ions in water
- complex ions do not: $[Fe(CN)_6]^{4-}$ of $K_4[Fe(CN)_6]$ does not split into Fe^{2+} and CN^- ions

Werner (1866–1919): life and work

- born 12 December 1866 in Mülhouse, a small community in the French province of Alsace
- studied chemistry first in Karlsruhe (Germany), then in Zurich (Switzerland); his 1890 doctoral thesis explained property differences among certain nitrogen-containing organic substances by isomerism
- extended van't Hoff's theory of the tetrahedral carbon atom and adapted it for nitrogen
- used physical measurements to show optical and electrical differences between complex compounds; he was the first to discover optical activity in certain coordination compounds
- became a full professor at the Technische Hochschule, Zurich, in 1895, aged 29; a chemist and educationist
- his theory of coordination compounds, with its revolutionary ideas on how atoms and molecules link together, was worked out in only three years, 1890–1893; he spent the rest of his career gathering the experimental support to confirm it
- his work on how atoms are linked, and on coordination theory, earned him the 1913 Nobel Prize, the first won by a Swiss chemist

5.2 Definitions of Some Important Terms Pertaining to Coordination Compounds

(a) Coordination entity

- a central metal atom or ion bonded to a fixed number of ions or molecules
- e.g. $[CoCl_3(NH_3)_3]$: cobalt ion surrounded by three NH_3 molecules and three Cl^- ions; also $[Ni(CO)_4]$, $[PtCl_2(NH_3)_2]$, $[Fe(CN)_6]^{4-}$, $[Co(NH_3)_6]^{3+}$

(b) Central atom/ion

- the atom or ion in a coordination entity around which a fixed number of ions/groups are bound in a definite geometry
- in $[NiCl_2(H_2O)_4]$, $[CoCl(NH_3)_5]^{2+}$ and $[Fe(CN)_6]^{3-}$ they are Ni^{2+}, Co^{3+} and Fe^{3+}; central atoms/ions are also called **Lewis acids**

(c) Ligands

- **ligands**: ions or molecules bound to the central atom/ion; from simple ions (Cl^-) and small molecules (H_2O, NH_3) to larger ones ($H_2NCH_2CH_2NH_2$, $N(CH_2CH_2NH_2)_3$) and even macromolecules such as proteins
- **unidentate**: bound through one donor atom (Cl^-, H_2O, NH_3); **didentate**: two donor atoms ($H_2NCH_2CH_2NH_2$, ethane-1,2-diamine; $C_2O_4^{2-}$, oxalate); **polydentate**: several donor atoms in one ligand ($N(CH_2CH_2NH_2)_3$)
- ethylenediaminetetraacetate ($EDTA^{4-}$) is an important **hexadentate** ligand, binding through two N and four O atoms
- **chelate ligand**: a di- or polydentate ligand using two or more donor atoms at once to bind one metal ion; the number of such binding groups is its **denticity**
- **chelate complexes** tend to be more stable than similar complexes of unidentate ligands
- **ambidentate ligand**: has two different donor atoms and can bind through either; e.g. NO_2^- (through N or O) and SCN^- (through S or N)

(d) Coordination number (CN)

- the number of ligand donor atoms directly bonded to the metal
- $[PtCl_6]^{2-}$: CN of Pt = 6; $[Ni(NH_3)_4]^{2+}$: CN of Ni = 4
- $[Fe(C_2O_4)_3]^{3-}$ and $[Co(en)_3]^{3+}$: CN = 6 for both Fe and Co, because $C_2O_4^{2-}$ and en (ethane-1,2-diamine) are didentate

Only the σ bonds a ligand forms with the central atom/ion count towards the coordination number; any π bonds between them are not counted.

(e) Coordination sphere

- the central atom/ion together with the ligands on it, all written inside one square bracket
- ionisable groups written outside the bracket are **counter ions**; in $K_4[Fe(CN)_6]$, the sphere is $[Fe(CN)_6]^{4-}$ and the counter ion K^+

(f) Coordination polyhedron

- the shape traced out in space by the ligand atoms bonded directly to the central atom/ion
- commonest: octahedral ($[Co(NH_3)_6]^{3+}$), square planar ($[PtCl_4]^{2-}$) and tetrahedral ($[Ni(CO)_4]$); shapes in Figure 5.A

Figure 5.A Shapes of different coordination polyhedra. M represents the central atom/ion and L, a unidentate ligand.

- **octahedral**: six L around M, four in a square plane with one above and one below
- **tetrahedral**: four L at the corners of a tetrahedron
- **square planar**: four L at the corners of a square, in one plane with M
- **trigonal bipyramidal**: five L, three in a triangle round M with one above and one below
- **square pyramidal**: five L, four forming a square base with one at the apex

(g) Oxidation number of central atom

- the charge the central atom would carry if all ligands were removed together with the electron pairs they share with it
- shown as a Roman numeral in brackets after the name of the coordination entity; Cu in $[Cu(CN)_4]^{3-}$ is +1, written Cu(I)

(h) Homoleptic and heteroleptic

- **homoleptic**: metal bound to only one kind of donor group, e.g. $[Co(NH_3)_6]^{3+}$
- **heteroleptic**: metal bound to more than one kind, e.g. $[Co(NH_3)_4Cl_2]^+$

5.3 Nomenclature of Coordination Compounds

- naming matters because formulas and systematic names must be unambiguous, above all for isomers
- the formulas and names used for coordination entities are those recommended by IUPAC, the International Union of Pure and Applied Chemistry

5.4 Formulas of Mononuclear Coordination Entities

Purpose

- a formula is shorthand that conveys the basic constitution of a compound concisely
- **mononuclear** coordination entities have a single central metal atom
- note: the 2004 IUPAC draft recommends sorting ligands alphabetically, whatever their charge

Rules for writing formulas

- (i) central atom first; (ii) then ligands in alphabetical order, regardless of charge
- (iii) polydentate ligands are listed alphabetically too; an abbreviated ligand is placed by the first letter of its abbreviation
- (iv) the whole entity, charged or not, goes in square brackets; polyatomic ligands and ligand abbreviations go in parentheses
- (v) no space between ligands and metal inside the sphere; (vi) a charged entity written without its counter ion shows the charge outside the bracket as a right superscript, number before sign, e.g. $[Co(CN)_6]^{3-}$, $[Cr(H_2O)_6]^{3+}$; (vii) cation charge balances anion charge

5.5 Naming of Mononuclear Coordination Compounds

Additive nomenclature

- names follow additive nomenclature: the groups around the central atom are named as prefixes to it, with any needed multipliers
- note: the 2004 IUPAC draft recommends anionic ligand names end in –ido, so chloro becomes chlorido, etc.

Rules for naming (i)–(iii)

- (i) the cation is named first, whether the entity is positive or negative
- (ii) ligands come in alphabetical order before the central atom/ion (the reverse of formula writing)
- (iii) anionic ligands end in –o; neutral and cationic ligands keep their names except aqua (H_2O), ammine (NH_3), carbonyl (CO) and nitrosyl (NO); in the entity's formula these are enclosed in brackets ( )

Rules for naming (iv)–(vi)

- (iv) mono, di, tri, etc. give the number of each ligand; where a ligand name already contains a numerical prefix, bis, tris, tetrakis are used with the ligand in parentheses, e.g. $[NiCl_2(PPh_3)_2]$ is dichloridobis(triphenylphosphine)nickel(II)
- (v) the metal's oxidation state, in a cationic, anionic or neutral entity, is shown by a Roman numeral in parentheses
- (vi) in a complex cation the metal keeps the element's name (Co = cobalt, Pt = platinum); in a complex anion it takes the suffix –ate, e.g. Co in $[Co(SCN)_4]^{2-}$ is cobaltate; some metals use Latin names in anions, e.g. ferrate for Fe

(vii) A neutral complex molecule is named in the same way as a complex cation.

Naming in practice

1. $[Cr(NH_3)_3(H_2O)_3]Cl_3$:
triamminetriaquachromium(III) chloride

- the bracketed complex ion is a cation; ammine ligands precede aqua alphabetically
- three chloride ions in a neutral compound make the complex ion +3; the ligands are all neutral, so chromium's oxidation number equals the ion's charge, +3

2. $[Co(H_2NCH_2CH_2NH_2)_3]_2(SO_4)_3$: tris(ethane-1,2–diamine)cobalt(III) sulphate

- sulphate is the counter anion; three sulphates balance two complex cations, so each cation is +3
- ethane-1,2–diamine is neutral, so cobalt is +3; the numbers of cations and anions are never stated in the name of an ionic compound

3. $[Ag(NH_3)_2][Ag(CN)_2]$: diamminesilver(I)dicyanidoargentate(I). Note how the same metal ion is named differently in the cation and in the anion.

Worked example formulas from names

Name	Formula
(a) tetraammineaquachloridocobalt(III) chloride	$[Co(NH_3)_4(H_2O)Cl]Cl_2$
(b) potassium tetrahydroxidozincate(II)	$K_2[Zn(OH)_4]$
(c) potassium trioxalatoaluminate(III)	$K_3[Al(C_2O_4)_3]$
(d) dichloridobis(ethane-1,2-diamine)cobalt(III)	$[CoCl_2(en)_2]^+$
(e) tetracarbonylnickel(0)	$[Ni(CO)_4]$

Worked example IUPAC names from formulas

Formula	IUPAC name
(a) $[Pt(NH_3)_2Cl(NO_2)]$	diamminechloridonitrito-N-platinum(II)
(b) $K_3[Cr(C_2O_4)_3]$	potassium trioxalatochromate(III)
(c) $[CoCl_2(en)_2]Cl$	dichloridobis(ethane-1,2-diamine)cobalt(III) chloride
(d) $[Co(NH_3)_5(CO_3)]Cl$	pentaamminecarbonatocobalt(III) chloride
(e) $Hg[Co(SCN)_4]$	mercury (I) tetrathiocyanato-S-cobaltate(III)

Intext questions

Q1 Give the formulas of: (i) tetraamminediaquacobalt(III) chloride; (ii) potassium tetracyanidonickelate(II); (iii) tris(ethane–1,2–diamine) chromium(III) chloride; (iv) amminebromidochloridonitrito-N-platinate(II); (v) dichloridobis(ethane–1,2–diamine)platinum(IV) nitrate; (vi) iron(III) hexacyanidoferrate(II).

Q2 Give the IUPAC names of: (i) $[Co(NH_3)_6]Cl_3$; (ii) $[Co(NH_3)_5Cl]Cl_2$; (iii) $K_3[Fe(CN)_6]$; (iv) $K_3[Fe(C_2O_4)_3]$; (v) $K_2[PdCl_4]$; (vi) $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$.

5.6 Isomerism in Coordination Compounds

Isomers

- compounds sharing one chemical formula whose atoms are arranged differently; the different arrangement shows up as differences in physical or chemical properties
- **stereoisomers**: same formula and same bonds, different spatial arrangement
- **structural isomers**: different bonds

Two main types, each subdivided

- (a) **stereoisomerism**: (i) geometrical, (ii) optical
- (b) **structural isomerism**: (i) linkage, (ii) coordination, (iii) ionisation, (iv) solvate

5.7 Geometric Isomerism

Square planar and octahedral cases

- arises in heteroleptic complexes from different possible geometric arrangements of ligands; key examples have coordination numbers 4 and 6
- square planar $[MX_2L_2]$ (X, L unidentate): the two X can be next to each other (**cis**) or opposite (**trans**) (Figure 5.B)
- square planar MABXL (all unidentate) gives three isomers, two cis and one trans; try drawing them
- impossible for tetrahedral geometry, but octahedral $[MX_2L_4]$ can have the two X cis or trans (Figure 5.C)

Figure 5.B Geometrical isomers (cis and trans) of Pt $[NH_3)_2Cl_2]$

- square planar Pt: in cis the two Cl sit on adjacent corners (as do the two NH_3); in trans the two Cl sit on opposite corners

Figure 5.C Geometrical isomers (cis and trans) of $[Co(NH_3)_4Cl_2]^+$

- octahedral Co with four NH_3: in cis the two Cl occupy neighbouring positions; in trans they lie opposite each other, above and below

With didentate ligands

- geometrical isomerism also occurs with didentate ligands L–L (e.g. $NH_2CH_2CH_2NH_2$, en) in complexes $[MX_2(L–L)_2]$ (Figure 5.D)

Figure 5.D Geometrical isomers (cis and trans) of $[CoCl_2(en)_2]$

- octahedral Co with two en rings: cis has the two Cl adjacent; trans has them on opposite sides, with both en rings in the plane between

fac and mer isomers

- octahedral entities $[Ma_3b_3]$, such as $[Co(NH_3)_3(NO_2)_3]$, show another kind (Figure 5.E)
- **facial (fac)**: three identical donor atoms on the corners of one octahedral face
- **meridional (mer)**: the three lie around a meridian of the octahedron

Figure 5.E The facial (fac) and meridional (mer) isomers of $[Co(NH_3)_3(NO_2)_3]$

- fac: the three NO_2 groups cluster on one triangular face, the three NH_3 on the opposite face
- mer: each set of three lies in one plane with the metal, two of the three opposite each other and the third between them

Worked example tetrahedral complexes

Why can tetrahedral complexes with two different kinds of unidentate ligand on the central metal ion not show geometrical isomerism?

- in a tetrahedron every unidentate ligand has the same relative position with respect to the others, so no distinct geometrical arrangements are possible

5.8 Optical Isomerism

Enantiomers

- optical isomers are non-superimposable mirror images, called **enantiomers**; molecules or ions that cannot be superimposed are **chiral**
- the two forms are **dextro (d)** and **laevo (l)**, according to the direction they turn plane-polarised light in a polarimeter (d to the right, l to the left)
- common in octahedral complexes of didentate ligands (Figure 5.F); in $[PtCl_2(en)_2]^{2+}$ only the cis isomer is optically active (Figure 5.G)

Figure 5.F Optical isomers (d and l) of $[Co(en)_3]^{3+}$

- two octahedral Co ions, each wrapped by three en rings, drawn as mirror images that cannot be superimposed

Figure 5.G Optical isomers (d and l) of cis-$[PtCl_2(en)_2]^{2+}$

- two mirror-image forms of the cis isomer, each with the two Cl adjacent and two en rings, not superimposable on each other

Worked example geometrical isomers

Sketch the geometrical isomers of $[Fe(NH_3)_2(CN)_4]^-$.

- both are octahedral about Fe, with four CN^- and two NH_3
- **cis**: the two NH_3 occupy neighbouring positions
- **trans**: the two NH_3 lie opposite each other, above and below the plane of the four CN^-

Worked example which is chiral?

Which of these is chiral (optically active): (a) cis-$[CrCl_2(ox)_2]^{3-}$ or (b) trans-$[CrCl_2(ox)_2]^{3-}$?

- (a) cis: the two Cl are adjacent and the two ox rings are arranged unsymmetrically around Cr
- (b) trans: the two Cl lie opposite each other, with both ox rings in the plane between, giving a mirror plane
- so (a) **cis**-$[CrCl_2(ox)_2]^{3-}$ is the chiral (optically active) one

5.9 Linkage Isomerism

- occurs in coordination compounds containing an **ambidentate** ligand
- thiocyanate, NCS^-, may bind through nitrogen (M–NCS) or through sulphur (M–SCN)
- Jørgensen found this in $[Co(NH_3)_5(NO_2)]Cl_2$: a **red** form with nitrite bound through oxygen (–ONO) and a **yellow** form with nitrite bound through nitrogen ($–NO_2$)

5.10 Coordination Isomerism

- occurs when ligands swap places between the cationic and the anionic entity, each built on a different metal ion
- in $[Co(NH_3)_6][Cr(CN)_6]$, NH_3 is on Co^{3+} and CN^- on Cr^{3+}; in its coordination isomer $[Cr(NH_3)_6][Co(CN)_6]$, NH_3 is on Cr^{3+} and CN^- on Co^{3+}

5.11 Ionisation Isomerism

- occurs when a complex salt's counter ion could itself act as a ligand: it can push out a ligand, and the ligand pushed out becomes the counter ion
- example: the ionisation isomers $[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$

5.12 Solvate Isomerism

- called 'hydrate isomerism' when the solvent is water; similar to ionisation isomerism
- solvate isomers differ in whether a solvent molecule is bonded directly to the metal ion or is merely present as free solvent in the crystal lattice
- example: the aqua complex $[Cr(H_2O)_6]Cl_3$ (violet) and its solvate isomer $[Cr(H_2O)_5Cl]Cl_2.H_2O$ (grey-green)

Intext questions

Q3 State the kinds of isomerism shown by these complexes and draw the isomers: (i) $K[Cr(H_2O)_2(C_2O_4)_2]$; (ii) $[Co(en)_3]Cl_3$; (iii) $[Co(NH_3)_5(NO_2)](NO_3)_2$; (iv) $[Pt(NH_3)(H_2O)Cl_2]$.

Q4 Show that $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5(SO_4)]Cl$ are ionisation isomers.

5.13 Bonding in Coordination Compounds

What Werner could not explain

- Werner first described bonding in coordination compounds, but his theory left basic questions open:
- (i) why only certain elements form coordination compounds so remarkably; (ii) why the bonds are directional
- (iii) where the distinctive magnetic and optical behaviour of coordination compounds comes from

Approaches

- Valence Bond Theory (VBT), Crystal Field Theory (CFT), Ligand Field Theory (LFT) and Molecular Orbital Theory (MOT)
- here only an elementary treatment of VBT and CFT is given

5.14 Valence Bond Theory

- under the ligands' influence the metal atom or ion hybridises its (n–1)d, ns, np or ns, np, nd orbitals into a set of equivalent orbitals of definite geometry: octahedral, tetrahedral, square planar and so on (Table 5.B)
- these hybrid orbitals overlap with ligand orbitals that donate electron pairs for bonding

Table 5.B Number of Orbitals and Types of Hybridisations

Coordination number	Type of hybridisation	Spatial arrangement of hybrid orbitals
4	sp^3	Tetrahedral
4	dsp^2	Square planar
5	sp^3d	Trigonal bipyramidal
6	sp^3d^2	Octahedral
6	d^2sp^3	Octahedral

Valence bond theory usually allows the geometry of a complex to be predicted from its magnetic behaviour.

$[Co(NH_3)_6]^{3+}$: inner orbital

- diamagnetic and octahedral; Co is +3, configuration $3d^6$ (hybridisation in Figure 5.H)
- six electron pairs, one from each NH_3, fill the six d^2sp^3 hybrids: octahedral, and diamagnetic with no unpaired electron
- it uses the inner 3d orbitals, so it is an **inner orbital**, **low spin** or **spin paired** complex

Figure 5.H

	3d	4s	4p
Co^{3+} ion	↑↓ ↑ ↑ ↑ ↑	–	– – –
d^2sp^3 hybridised	↑↓ ↑↓ ↑↓ | six empty d^2sp^3 hybrids		
$[Co(NH_3)_6]^{3+}$	↑↓ ↑↓ ↑↓ | six pairs from six NH_3		

$[CoF_6]^{3-}$: outer orbital

- paramagnetic and octahedral; hybridisation uses the outer 4d orbitals (sp^3d^2) (Figure 5.I)
- so it is an **outer orbital**, **high spin** or **spin free** complex

Figure 5.I

	3d	4s, 4p, 4d
Co^{3+} ion	↑↓ ↑ ↑ ↑ ↑	all empty
sp^3d^2 hybridised	↑↓ ↑ ↑ ↑ ↑	six empty sp^3d^2 hybrids; two 4d left
$[CoF_6]^{3-}$	↑↓ ↑ ↑ ↑ ↑	six pairs from six F^-

Tetrahedral: $[NiCl_4]^{2-}$

- one s and three p orbitals hybridise into four equivalent, tetrahedrally directed orbitals
- Ni is +2, configuration $3d^8$ (Figure 5.J); each Cl^- donates an electron pair
- two unpaired electrons remain, so it is paramagnetic
- $[Ni(CO)_4]$ is also tetrahedral but diamagnetic: Ni is in the zero oxidation state with no unpaired electron

Figure 5.J

	3d	4s, 4p
Ni^{2+} ion	↑↓ ↑↓ ↑↓ ↑ ↑	empty
sp^3 hybridised	↑↓ ↑↓ ↑↓ ↑ ↑	four empty sp^3 hybrids
$[NiCl_4]^{2-}$ (high spin)	↑↓ ↑↓ ↑↓ ↑ ↑	four pairs from 4 Cl^-

Square planar: $[Ni(CN)_4]^{2-}$

- square planar complexes use dsp^2 hybridisation; Ni is +2, configuration $3d^8$ (Figure 5.K)
- each hybrid orbital takes an electron pair from a cyanide ion; with no unpaired electron the compound is diamagnetic
- hybrid orbitals do not actually exist: hybridisation is a mathematical manipulation of the wave equation for the atomic orbitals involved

Figure 5.K

	3d	4s, 4p
Ni^{2+} ion	↑↓ ↑↓ ↑↓ ↑ ↑	empty
dsp^2 hybridised	↑↓ ↑↓ ↑↓ ↑↓ (one 3d emptied)	four empty dsp^2 hybrids; one 4p left
$[Ni(CN)_4]^{2-}$ (low spin)	↑↓ ↑↓ ↑↓ ↑↓	four pairs from 4 CN^-

5.15 Magnetic Properties of Coordination Compounds

What magnetic data reveal

- magnetic moments are measured by magnetic susceptibility experiments; they give the number of unpaired electrons and hence the structures of metal complexes
- up to three d electrons (Ti^{3+} d^1, V^{3+} d^2, Cr^{3+} d^3): two d orbitals stay vacant for octahedral hybridisation with 4s and 4p, and free ion and complex behave alike magnetically
- with more than three 3d electrons, Hund's rule means the needed pair of 3d orbitals is not directly free; for d^4 (Cr^{2+}, Mn^{3+}), d^5 (Mn^{2+}, Fe^{3+}), d^6 (Fe^{2+}, Co^{3+}) a vacant pair appears only by pairing 3d electrons, leaving two, one and zero unpaired

Apparent anomalies

- maximum spin pairing fits the data in many cases, especially d^6 ions; d^4 and d^5 species are complicated
- $[Mn(CN)_6]^{3-}$: two unpaired electrons; $[MnCl_6]^{3-}$: four. $[Fe(CN)_6]^{3-}$: one; $[FeF_6]^{3-}$: five. $[CoF_6]^{3-}$: paramagnetic with four; $[Co(C_2O_4)_3]^{3-}$: diamagnetic

VBT explanation: inner vs outer orbital

- **inner orbital** (d^2sp^3): $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[Co(C_2O_4)_3]^{3-}$; the first two paramagnetic, the last diamagnetic
- **outer orbital** (sp^3d^2): $[MnCl_6]^{3-}$, $[FeF_6]^{3-}$ and $[CoF_6]^{3-}$; paramagnetic, with four, five and four unpaired electrons

Worked example geometry from magnetic moment

$[MnBr_4]^{2-}$ has a spin-only magnetic moment of 5.9 BM. What is its geometry?

- Mn^{2+} has coordination number 4 here, so the ion is either tetrahedral (sp^3) or square planar (dsp^2)
- 5.9 BM means five unpaired d electrons, so it must be **tetrahedral**, not square planar

5.16 Limitations of Valence Bond Theory

VBT explains formation, structure and magnetism largely, but:

- (i) it rests on several assumptions
- (ii) it gives no quantitative account of magnetic data
- (iii) the colours of coordination compounds are left unexplained
- (iv) it gives no quantitative account of thermodynamic or kinetic stability
- (v) it cannot predict exactly whether 4-coordinate complexes are tetrahedral or square planar
- (vi) it does not distinguish weak from strong ligands

5.17 Crystal Field Theory

An electrostatic model

- CFT treats the metal–ligand bond as ionic, arising purely from electrostatic interaction between metal ion and ligand
- ligands are point charges (anions) or point dipoles (neutral molecules)

Lifting the degeneracy

- in an isolated gaseous metal atom/ion, the five d orbitals have the same energy (degenerate); a spherically symmetric field of negative charge keeps them so
- a field from ligands (anions, or the negative ends of dipoles such as NH_3 and H_2O) is not symmetric, so the d orbitals **split**; the pattern depends on the crystal field

(a) Splitting in octahedral coordination entities

- six ligands surround the metal; electrons in the metal d orbitals are repelled by the ligand electrons (negative charges), more so for orbitals pointing at the ligands
- $d_{x^2-y^2}$ and d_{z^2} point along the axes towards the ligands: more repulsion, raised in energy
- d_{xy}, d_{yz}, d_{xz} point between the axes: lowered relative to the average energy in a spherical field
- the result: three lower-energy orbitals, the $\mathbf{t_{2g}}$ set, and two higher-energy orbitals, the $\mathbf{e_g}$ set

Crystal field splitting, Δ_o

- splitting of the degenerate levels by ligands in a definite geometry is **crystal field splitting**; the energy gap is Δ_o (o for octahedral) (Figure 5.L)
- the two e_g orbitals rise by $(3/5)\Delta_o$; the three t_{2g} fall by $(2/5)\Delta_o$
- Δ_o depends on the field from the ligand and on the charge of the metal ion: strong-field ligands give large splitting, weak-field ligands small

Figure 5.L d orbital splitting in an octahedral crystal field

- energy levels: free ion (five degenerate d), then the average level in a spherical field, then the octahedral field
- in the octahedral field, e_g ($d_{x^2-y^2}$, d_{z^2}) lies $+(3/5)\Delta_o$ above the average and t_{2g} (d_{xy}, d_{yz}, d_{xz}) lies $-(2/5)\Delta_o$ below it

Spectrochemical series

- ligands in order of increasing field strength:

$$I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$$

- determined by experiment, from the light absorbed by complexes of different ligands

Filling the d orbitals (octahedral)

- d^1: the electron goes into a lower t_{2g} orbital; d^2, d^3: electrons occupy t_{2g} orbitals singly (Hund's rule)
- d^4: the fourth electron either pairs in t_{2g}, or avoids the pairing energy by going into e_g
- which happens depends on Δ_o versus the **pairing energy** *P*, the energy needed to pair electrons in one orbital

Weak vs strong field

- (i) $\Delta_o < P$: fourth electron enters e_g, giving $t_{2g}^3 e_g^1$; ligands like this are called **weak field ligands**; the complexes they give are **high spin**
- (ii) $\Delta_o > P$: fourth electron pairs in t_{2g}, giving $t_{2g}^4 e_g^0$; ligands causing this are **strong field ligands**, giving **low spin** complexes
- calculations show d^4 to d^7 entities are more stable in strong field than weak field cases

(b) Splitting in tetrahedral coordination entities

- the d orbital splitting is **inverted** and **smaller** than in an octahedral field (Figure 5.M)
- for the same metal, ligands and metal–ligand distances, $\Delta_t = (4/9)\Delta_o$
- the splitting is too small to force pairing, so low spin configurations are seldom seen
- the 'g' subscript is used for octahedral and square planar complexes, which have a centre of symmetry; tetrahedral complexes lack it, so their levels carry no 'g'

Figure 5.M d orbital splitting in a tetrahedral crystal field.

- from the free-ion d orbitals to their average energy in a spherical field, then splitting in the tetrahedral field
- t_2 (d_{xy}, d_{yz}, d_{xz}) rises by $(2/5)\Delta_t$; e ($d_{x^2-y^2}$, d_{z^2}) falls by $(3/5)\Delta_t$

5.18 Colour in Coordination Compounds

- a very distinctive feature of transition metal complexes is their wide range of colours (previous Unit)
- part of the visible spectrum is removed from white light passing through the sample, so the light emerging is no longer white
- the complex shows the **complementary** colour, the colour of the wavelengths left over: absorbing green light makes it look red; see Table 5.C

TABLE 5.C **Relationship between the Wavelength of Light absorbed and the Colour observed in some Coordination Entities**

Coordination entity	Wavelength of light absorbed (nm)	Colour of light absorbed	Colour of coordination entity
$[CoCl(NH_3)_5]^{2+}$	535	Yellow	Violet
$[Co(NH_3)_5(H_2O)]^{3+}$	500	Blue Green	Red
$[Co(NH_3)_6]^{3+}$	475	Blue	Yellow Orange
$[Co(CN)_6]^{3-}$	310	Ultraviolet (outside the visible region)	Pale Yellow
$[Cu(H_2O)_4]^{2+}$	600	Red	Blue
$[Ti(H_2O)_6]^{3+}$	498	Blue Green	Violet

Colour by crystal field theory: $[Ti(H_2O)_6]^{3+}$

- a violet octahedral complex; Ti^{3+} is $3d^1$, and in the ground state its single electron is in the t_{2g} level, with the e_g level empty above it
- absorbing light of blue-green energy promotes the electron from t_{2g} to e_g ($t_{2g}^1 e_g^0 \rightarrow t_{2g}^0 e_g^1$), so the complex looks violet (Figure 5.N)
- CFT thus attributes the colour of coordination compounds to **d–d transitions** of the electron

Figure 5.N Transition of an electron in

- energy diagram: in the ground state one electron sits in the lower t_{2g} level with e_g empty
- absorbing a photon ($h\nu$) raises it to e_g, giving the excited state with t_{2g} empty

No ligand, no colour

- without ligands there is no crystal field splitting, so the substance is colourless
- heating $[Ti(H_2O)_6]Cl_3$ to remove the water leaves it colourless; anhydrous $CuSO_4$ is white, but $CuSO_4.5H_2O$ is blue

The ligand changes the colour: nickel(II) with en

- dissolving nickel(II) chloride in water gives $[Ni(H_2O)_6]^{2+}$; adding ethane-1,2-diamine (en) step by step in en:Ni ratios 1:1, 2:1, 3:1 gives these reactions and colour changes (Figure 5.O):

$[Ni(H_2O)_6]^{2+}(aq)$ green + $en(aq) = [Ni(H_2O)_4(en)]^{2+}(aq)$ pale blue + $2H_2O$

$[Ni(H_2O)_4(en)]^{2+}(aq) + en(aq) = [Ni(H_2O)_2(en)_2]^{2+}(aq)$ blue/purple + $2H_2O$

$[Ni(H_2O)_2(en)_2]^{2+}(aq) + en(aq) = [Ni(en)_3]^{2+}(aq)$ violet + $2H_2O$

Figure 5.O Aqueous solutions of complexes of nickel(II) with an increasing number of ethane-1, 2-diamine ligands.

- four tubes in a rack, from left to right $[Ni(H_2O)_6]^{2+}(aq)$, $[Ni(H_2O)_4en]^{2+}(aq)$, $[Ni(H_2O)_2en_2]^{2+}(aq)$ and $[Ni(en)_3]^{2+}(aq)$, each showing the colour given in the reactions above

Colour of some gem stones: ruby

- colours from electronic transitions within the d orbitals of transition metal ions are common in everyday life
- **ruby** (Figure 5.P a) is aluminium oxide (Al_2O_3) with about 0.5–1% Cr^{3+} ions (d^3) scattered at random in sites normally held by Al^{3+}
- these Cr(III) centres can be seen as octahedral Cr(III) complexes set in the alumina lattice; their d–d transitions produce the colour

Emerald

- in emerald (Figure 5.P b), Cr^{3+} ions sit in octahedral sites of the mineral beryl ($Be_3Al_2Si_6O_{18}$)
- the absorption bands seen in ruby move to longer wavelengths, yellow-red and blue, so emerald transmits light in the green region

Figure 5.P (a) Ruby: this gemstone was found in marble from Mogok, Myanmar; (b) Emerald: this gemstone was found in Muzo, Columbia.

- (a) a ruby crystal embedded in marble; (b) an emerald crystal

5.19 Limitations of Crystal Field Theory

Successes and weaknesses

- to a large extent the crystal field model accounts for how coordination compounds form, their structures, their colours and their magnetism
- but treating ligands as point charges implies anionic ligands should split most, yet they actually sit at the low end of the spectrochemical series
- it also leaves out the covalent character of the bond between ligand and central atom; these weaknesses are addressed by ligand field theory (LFT) and molecular orbital theory, beyond the present scope

5.20 Bonding in Metal Carbonyls

Homoleptic carbonyls and their shapes

- compounds with only carbonyl ligands, formed by most transition metals, with simple, well-defined structures
- tetracarbonylnickel(0): tetrahedral; pentacarbonyliron(0): trigonal bipyramidal; hexacarbonylchromium(0): octahedral
- decacarbonyldimanganese(0): two square pyramidal $Mn(CO)_5$ units joined by an Mn–Mn bond; octacarbonyldicobalt(0): a Co–Co bond bridged by two CO groups (Figure 5.Q)

Intext questions

Q5 Use valence bond theory to explain why square planar $[Ni(CN)_4]^{2-}$ is diamagnetic while tetrahedral $[NiCl_4]^{2-}$ is paramagnetic.

Q6 Both $[NiCl_4]^{2-}$ and $[Ni(CO)_4]$ are tetrahedral, yet the first is paramagnetic and the second diamagnetic. Why?

Q7 Explain why $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic but $[Fe(CN)_6]^{3-}$ only weakly so.

Q8 Explain why $[Co(NH_3)_6]^{3+}$ is an inner orbital complex and $[Ni(NH_3)_6]^{2+}$ an outer orbital complex.

Q9 How many unpaired electrons does the square planar $[Pt(CN)_4]^{2-}$ ion have?

Q10 Using crystal field theory, explain why the hexaaqua manganese(II) ion has five unpaired electrons but the hexacyano ion only one.

Figure 5.Q Structures of some representative homoleptic metal carbonyls.

- **$Ni(CO)_4$**: tetrahedral, four CO around Ni
- **$Fe(CO)_5$**: trigonal bipyramidal, three CO in a triangle and one above, one below
- **$Cr(CO)_6$**: octahedral, six CO around Cr
- **$[Mn_2(CO)_{10}]$**: two $Mn(CO)_5$ units joined by an Mn–Mn bond
- **$[Co_2(CO)_8]$**: two $Co(CO)_3$ units joined by a Co–Co bond, bridged by two CO groups

Synergic bonding

- the metal–carbon bond in metal carbonyls has both σ and π character
- **M–C σ bond**: the lone pair on the carbonyl carbon is donated into a vacant metal orbital
- **M–C π bond**: a filled metal d orbital donates an electron pair into the vacant antibonding π* orbital of CO
- this metal-to-ligand bonding sets up a **synergic** effect that strengthens the CO–metal bond (Figure 5.R)

Figure 5.R Example of synergic bonding interactions in a carbonyl complex.

- the σ-type donation from the C lone pair into an empty metal orbital, and the π-type back-donation from a filled metal d orbital into an empty π* orbital of CO

5.21 Importance and Applications of Coordination Compounds

Coordination compounds occur throughout the mineral, plant and animal kingdoms, and do many jobs: in chemical analysis, metallurgy, living systems, industry and medicine.

Analysis

- widely used in qualitative and quantitative analysis: metal ions give familiar colour reactions with ligands (chelating ligands especially) by forming coordination entities, and these reactions are how the ions are detected and measured, by classical and instrumental methods alike
- reagents include EDTA, DMG (dimethylglyoxime), α-nitroso-β-naphthol, cupron, etc.
- water hardness is found by simple titration with Na_2EDTA: Ca^{2+} and Mg^{2+} form stable EDTA complexes, and differing stability constants allow each to be estimated selectively

Metallurgy

- some extraction processes, e.g. for silver and gold, rely on complex formation
- gold combines with cyanide in the presence of oxygen and water to give $[Au(CN)_2]^-$ in solution; adding zinc recovers metallic gold
- metals can likewise be purified by forming and then decomposing their coordination compounds

Purification and biology

- impure nickel is converted to $[Ni(CO)_4]$, which decomposes to pure nickel
- chlorophyll, which drives photosynthesis, is a coordination compound of magnesium; haemoglobin, blood's red oxygen carrier, is one of iron; vitamin B_{12} (cyanocobalamine), which guards against pernicious anaemia, is one of cobalt
- enzymes such as carboxypeptidase A and carbonic anhydrase (biological catalysts) also contain coordinated metal ions

Industry and photography

- catalysts for many industrial processes, e.g. the rhodium complex $[(Ph_3P)_3RhCl]$, Wilkinson catalyst, for hydrogenating alkenes
- plating from $[Ag(CN)_2]^-$ or $[Au(CN)_2]^-$ solutions gives far smoother, more even silver or gold coatings than plating from solutions of the simple ions
- black-and-white film, once developed, is fixed by washing in hypo; the hypo takes up the AgBr that was not decomposed, as the complex ion $[Ag(S_2O_3)_2]^{3-}$

Medicine: chelate therapy

- growing interest in medicinal chemistry, e.g. treating problems from toxic levels of metals in plant/animal systems
- the chelating ligands D-penicillamine and desferrioxamine B clear surplus copper and iron by binding them into coordination compounds; EDTA treats lead poisoning
- some platinum coordination compounds, such as cis-platin and related compounds, effectively inhibit tumour growth

Summary

The field

- an important and demanding area of modern inorganic chemistry
- over the last fifty years it has yielded new concepts and models of bonding and structure, breakthroughs in chemical industry, and vital insight into key components of living systems

Werner

- made the first systematic attempt to explain the formation, reactions, structure and bonding of coordination compounds
- his two linkages, primary and secondary, are now recognised as ionisable (ionic) and non-ionisable (covalent) bonds; using isomerism he predicted the shapes of many coordination entities

Valence Bond Theory

- explains fairly well the formation, magnetic behaviour and shapes of coordination compounds
- but gives no quantitative account of magnetism and says nothing about optical properties

Crystal Field Theory

- based on how different crystal fields (from ligands treated as point charges) affect the degeneracy of the central metal's d orbital energies

CFT, continued

- d orbital splitting gives different electron arrangements in strong and weak fields
- it puts numbers on the gaps between orbitals, on magnetic moments, and on spectral and stability quantities
- but treating ligands as point charges raises many theoretical difficulties

Carbonyls and applications

- the metal–carbon bond in carbonyls has σ character (ligand to metal) and π character (metal to ligand); this unique synergic bonding stabilises them
- coordination compounds give key insight into how vital biological components work and are structured, and are widely used in metallurgy and in analytical and medicinal chemistry

Exercises

E1 Explain bonding in coordination compounds using Werner's postulates.

E2 Solutions of $FeSO_4$ and $(NH_4)_2SO_4$ mixed 1:1 by moles still respond to the test for Fe^{2+}. Yet $CuSO_4$ solution mixed 1:4 by moles with aqueous ammonia gives no test for Cu^{2+}. Account for the difference.

E3 Using two examples for each, explain these terms: coordination entity; ligand; coordination number; coordination polyhedron; homoleptic; heteroleptic.

E4 Explain what unidentate, didentate and ambidentate ligands are, with two examples of each kind.

E5 State the oxidation number of the metal in: (i) $[Co(H_2O)(CN)(en)_2]^{2+}$; (ii) $[CoBr_2(en)_2]^{+}$; (iii) $[PtCl_4]^{2-}$; (iv) $K_3[Fe(CN)_6]$; (v) $[Cr(NH_3)_3Cl_3]$.

E6 Following IUPAC norms, write formulas for: (i) tetrahydroxidozincate(II); (ii) potassium tetrachloridopalladate(II); (iii) diamminedichloridoplatinum(II); (iv) potassium tetracyanidonickelate(II); (v) pentaamminenitrito-O-cobalt(III); (vi) hexaamminecobalt(III) sulphate; (vii) potassium tri(oxalato)chromate(III); (viii) hexaammineplatinum(IV); (ix) tetrabromidocuprate(II); (x) pentaamminenitrito-N-cobalt(III).

E7 Following IUPAC norms, give systematic names for: (i) $[Co(NH_3)_6]Cl_3$; (ii) $[Pt(NH_3)_2Cl(NH_2CH_3)]Cl$; (iii) $[Ti(H_2O)_6]^{3+}$; (iv) $[Co(NH_3)_4Cl(NO_2)]Cl$; (v) $[Mn(H_2O)_6]^{2+}$; (vi) $[NiCl_4]^{2-}$; (vii) $[Ni(NH_3)_6]Cl_2$; (viii) $[Co(en)_3]^{3+}$; (ix) $[Ni(CO)_4]$.

E8 List the types of isomerism possible in coordination compounds, with an example of each.

E9 How many geometrical isomers can exist for (i) $[Cr(C_2O_4)_3]^{3-}$; (ii) $[Co(NH_3)_3Cl_3]$?

E10 Draw the optical isomers of (i) $[Cr(C_2O_4)_3]^{3-}$; (ii) $[PtCl_2(en)_2]^{2+}$; (iii) $[Cr(NH_3)_2Cl_2(en)]^{+}$.

Exercises continued

E11 Draw every isomer, geometrical and optical, of (i) $[CoCl_2(en)_2]^+$; (ii) $[Co(NH_3)Cl(en)_2]^{2+}$; (iii) $[Co(NH_3)_2Cl_2(en)]^+$.

E12 Write all geometrical isomers of $[Pt(NH_3)(Br)(Cl)(py)]$. How many of them show optical isomerism?

E13 Aqueous copper sulphate, which is blue, forms (i) a green precipitate when aqueous potassium fluoride is added, and (ii) a bright green solution when aqueous potassium chloride is added. Explain both observations.

E14 What coordination entity forms when excess aqueous KCN is added to aqueous copper sulphate? Why does $H_2S(g)$ passed through this solution give no copper sulphide precipitate?

E15 Using valence bond theory, discuss the bonding in (i) $[Fe(CN)_6]^{4-}$; (ii) $[FeF_6]^{3-}$; (iii) $[Co(C_2O_4)_3]^{3-}$; (iv) $[CoF_6]^{3-}$.

E16 Draw a diagram showing how d orbitals split in an octahedral crystal field.

E17 What is the spectrochemical series? How does a weak field ligand differ from a strong field ligand?

E18 Define crystal field splitting energy. In a coordination entity, how does the magnitude of Δ_o determine which d orbital configuration is actually adopted?

E19 Why is $[Cr(NH_3)_6]^{3+}$ paramagnetic but $[Ni(CN)_4]^{2-}$ diamagnetic?

E20 Why is a $[Ni(H_2O)_6]^{2+}$ solution green while a $[Ni(CN)_4]^{2-}$ solution is colourless?

E21 Why do $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ have different colours in dilute solution?

E22 Discuss the bonding in metal carbonyls.

E23 For the central metal ion in each complex, state its oxidation state, its d orbital occupancy and its coordination number: (i) $K_3[Co(C_2O_4)_3]$; (ii) cis-$[CrCl_2(en)_2]Cl$; (iii) $(NH_4)_2[CoF_4]$; (iv) $[Mn(H_2O)_6]SO_4$.

E24 For each complex give the IUPAC name, oxidation state, electronic configuration, coordination number, stereochemistry and magnetic moment: (i) $K[Cr(H_2O)_2(C_2O_4)_2].3H_2O$; (ii) $[Co(NH_3)_5Cl]Cl_2$; (iii) $[CrCl_3(py)_3]$; (iv) $Cs[FeCl_4]$; (v) $K_4[Mn(CN)_6]$.

E25 Use crystal field theory to explain the violet colour of $[Ti(H_2O)_6]^{3+}$.

E26 What is the chelate effect? Give an example.

E27 With an example for each, discuss briefly the role of coordination compounds in (i) biological systems, (ii) medicinal chemistry, (iii) analytical chemistry and (iv) extraction/metallurgy of metals.

E28 How many ions does the complex $Co(NH_3)_6Cl_2$ give in solution? (i) 6 (ii) 4 (iii) 3 (iv) 2

Exercises continued

E29 Which of these ions has the highest magnetic moment? (i) $[Cr(H_2O)_6]^{3+}$ (ii) $[Fe(H_2O)_6]^{2+}$ (iii) $[Zn(H_2O)_6]^{2+}$

E30 Which complex is the most stable? (i) $[Fe(H_2O)_6]^{3+}$ (ii) $[Fe(NH_3)_6]^{3+}$ (iii) $[Fe(C_2O_4)_3]^{3-}$ (iv) $[FeCl_6]^{3-}$

E31 Put the wavelengths of visible absorption of $[Ni(NO_2)_6]^{4-}$, $[Ni(NH_3)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$ in the correct order.

Answers to some intext questions

Question	Answer
Q1	(i) $[Co(NH_3)_4(H_2O)_2]Cl_3$; (ii) $K_2[Ni(CN)_4]$; (iii) $[Cr(en)_3]Cl_3$; (iv) $[Pt(NH_3)BrCl(NO_2)]^-$; (v) $[PtCl_2(en)_2](NO_3)_2$; (vi) $Fe_4[Fe(CN)_6]_3$
Q2	(i) hexaamminecobalt(III) chloride; (ii) pentaamminechloridocobalt(III) chloride; (iii) potassium hexacyanidoferrate(III); (iv) potassium trioxalatoferrate(III); (v) potassium tetrachloridopalladate(II); (vi) diamminechlorido(methanamine)platinum(II) chloride
Q3	(i) geometrical (cis-, trans-) isomers, plus optical isomers for the cis form; (ii) two optical isomers; (iii) 10 possible isomers (hint: geometrical, ionisation and linkage isomers are possible); (iv) geometrical (cis-, trans-) isomers
Q4	ionisation isomers give different ions in water and so react differently with reagents: $[Co(NH_3)_5Br]SO_4 + Ba^{2+} \rightarrow BaSO_4(s)$, but $[Co(NH_3)_5SO_4]Br + Ba^{2+} \rightarrow$ no reaction; $[Co(NH_3)_5Br]SO_4 + Ag^+ \rightarrow$ no reaction, but $[Co(NH_3)_5SO_4]Br + Ag^+ \rightarrow AgBr(s)$
Q6	Ni is 0 in $Ni(CO)_4$ but +2 in $NiCl_4^{2-}$; with CO as ligand the unpaired d electrons of Ni pair up, whereas Cl^-, a weak ligand, cannot pair them
Q7	with CN^- (a strong ligand) the 3d electrons pair up, leaving one unpaired: d^2sp^3, inner orbital complex; with H_2O (a weak ligand) they stay unpaired: sp^3d^2, outer orbital complex with five unpaired electrons, strongly paramagnetic
Q8	with NH_3, the 3d electrons of Co(III) pair up, leaving two d orbitals empty for d^2sp^3: inner orbital $[Co(NH_3)_6]^{3+}$; in $Ni(NH_3)_6^{2+}$, Ni is +2 and d^8, so hybridisation is sp^3d^2: outer orbital complex
Q9	square planar means dsp^2; the unpaired electrons in the 5d orbitals pair up to empty one d orbital for dsp^2, so there is no unpaired electron